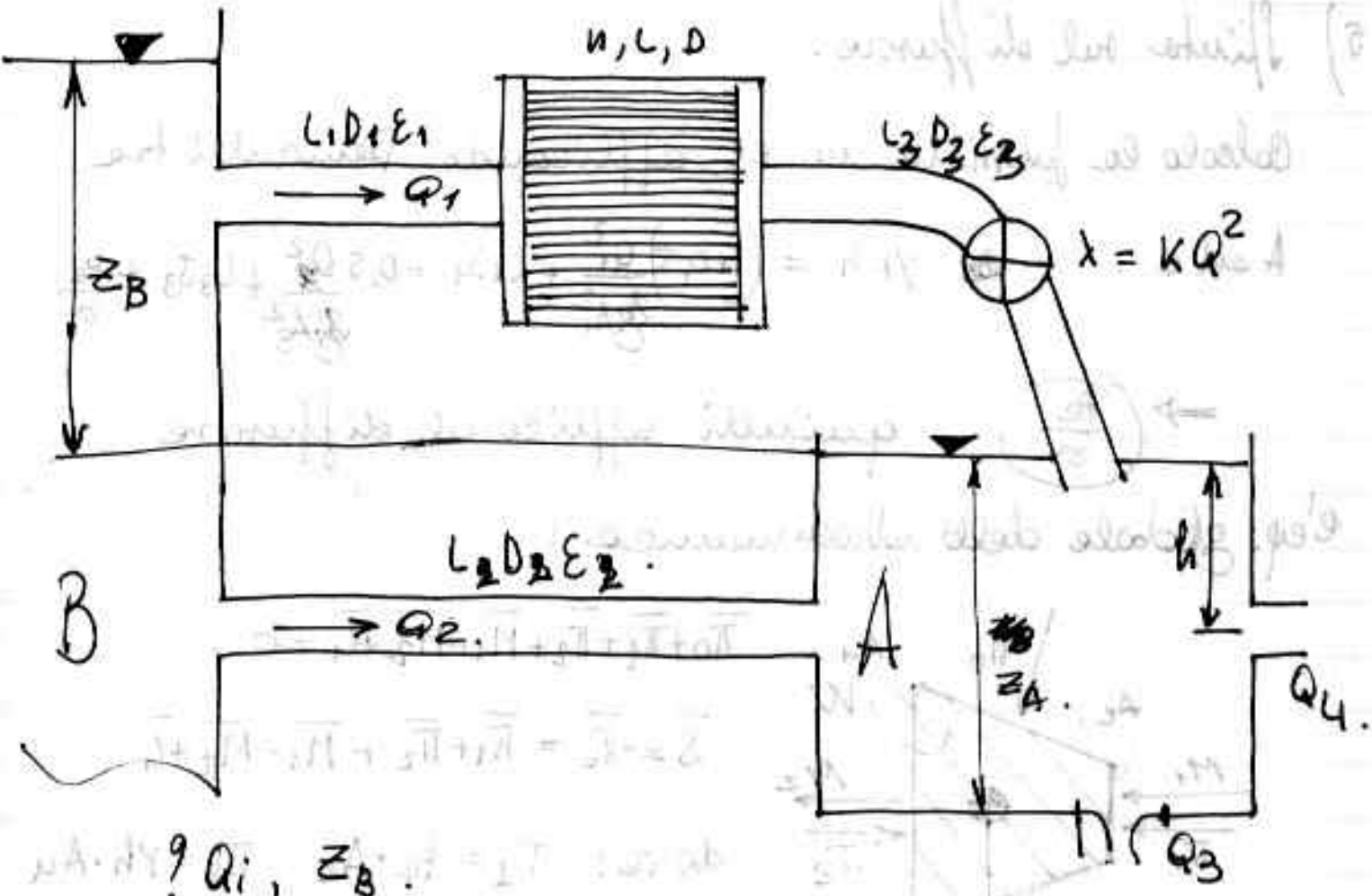


? Q_i, z_B.

Supposto che il serbatoio A mantenga il pelo libero a livello costante: $Q_3 = \mu_3 A_3 \sqrt{2g z_A}$ $\mu_3 \cong 0{,}61$

mentre $Q_4 = \mu_4 A_4 \sqrt{2g h}$ $\mu_4 \cong 0{,}98$.

per l'eq. di continuità si ha poi:

~~$Q_1+Q_2+Q_3+Q_4 = \text{cost}$~~ $Q_1+Q_2 = Q_3+Q_4$ $f(Q_1 Q_2)$

Applico B. tra B e A: attraverso 2

$$z_B = 0{,}5\frac{Q_2^2}{2gA_2} + L_2 J_2 + \frac{Q_1^2}{2gA_2}. \qquad f(z_B, Q_2)$$

dove $J_2 = \lambda_2 \frac{Q_2^2}{2gD^2A_2^2}$ $1/\sqrt{\lambda_2} = -2\log\left(\frac{1}{3{,}71}\frac{\varepsilon_2}{D_2}\right)$

Applico B. tra B e A fermando fra 1

$$z_B = J_1 l_1 + 0{,}5\frac{Q_1^2}{2gA_1} + \frac{Q_1^2}{2gC^2}\left(\frac{1}{A_1} - \frac{1}{A}\right)^2 + \sum J + 0{,}5\frac{Q_1^2}{2gA_3^2}$$

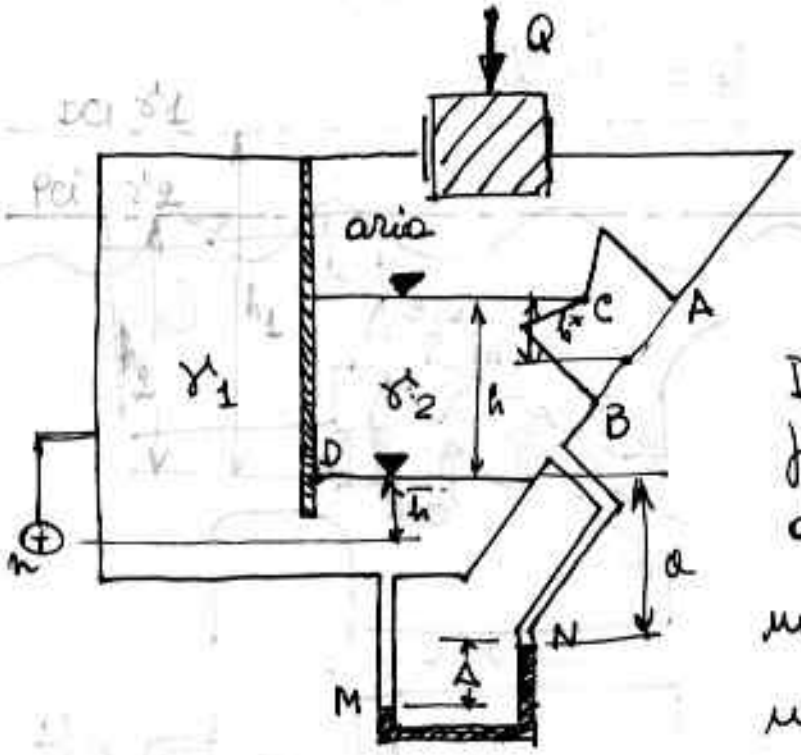
$$\sum J = 1\,\frac{Q_1^2}{n^2 g R}$$

$$+ kQ_1^2 + \frac{Q_1^2}{2gA_3^2}$$

$$f(Q_1, z_B)$$

Ho 3 eq. in 3 incognite che risolte forniscono

$$Q_1, Q_2, z_B .$$



Noti: geometria *, n, γ_1, γ_2

? Δ, spinta sulla sup. conica AB.

Detta Σ la superficie del pistone la pressione dell'aria vale: $\frac{Q}{\Sigma} = p_C$

mentre: $p_D = p_C + h\gamma_2$

mentre $n = \frac{\left(\frac{Q}{\Sigma} + h\gamma_2 + \bar{h}\gamma_1\right)}{10^4}$ (kg/cm²)

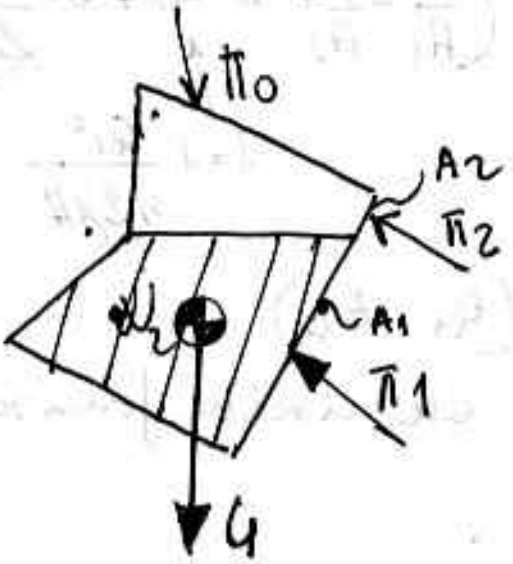
Il Pci di γ_2 dista da D di:

una quantità: dove $h_2 = \frac{p_D}{\gamma_2}$

mentre Pci di γ_1 dista, sempre da D di una altezza pari a

$$h_1 = \frac{p_D}{\gamma_1} \quad \text{dove } h_1 \neq h_2$$

$$\delta = (h_1+b)\frac{\gamma_1-\gamma_2}{\gamma_2} + \Delta\frac{\gamma_м-\gamma_2}{\gamma_2} \Rightarrow \Delta.$$



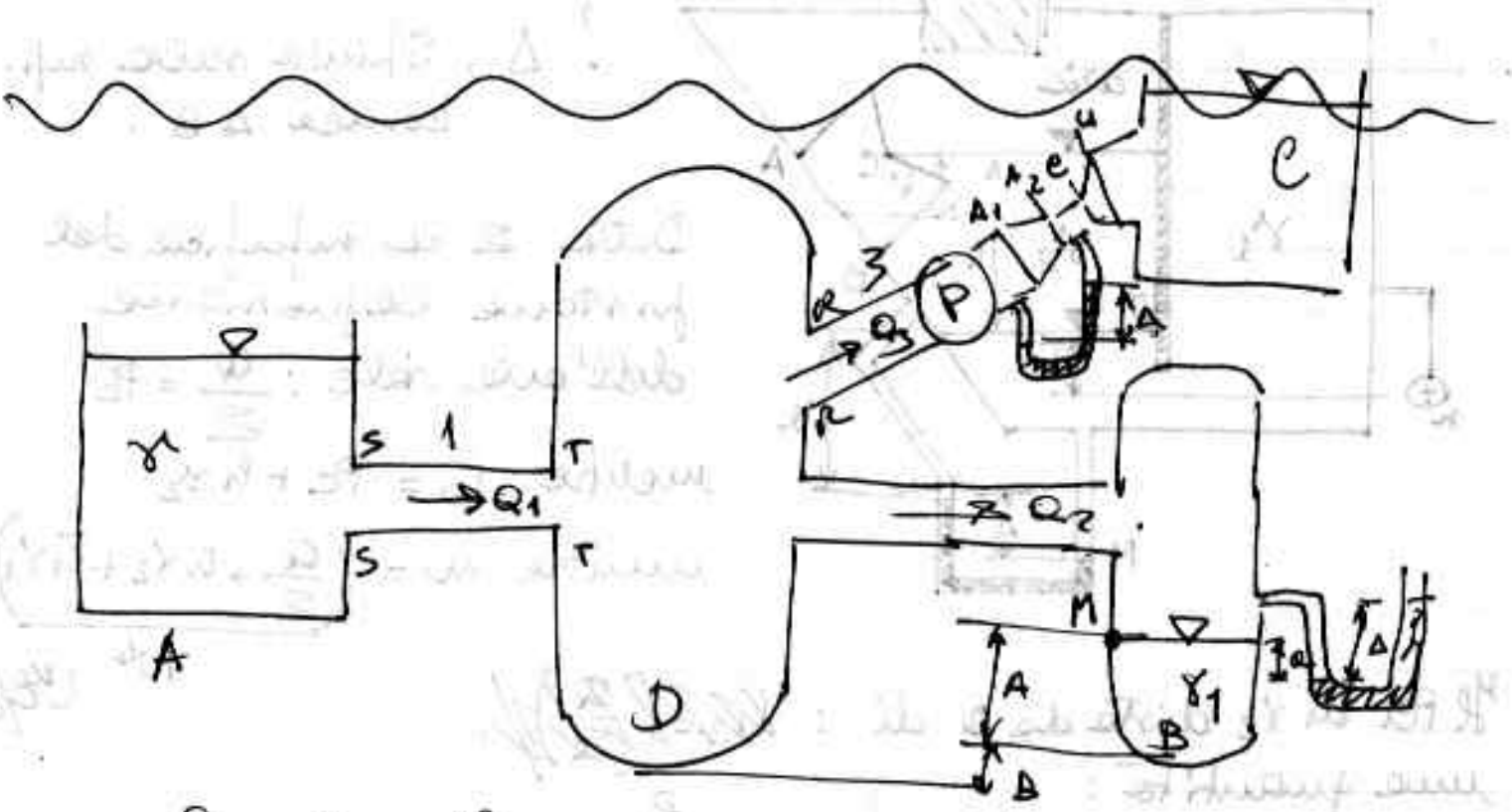
$$\bar{\Pi}_0 + \bar{\Pi}_1 + \bar{\Pi}_2 + \bar{G} = 0$$

$$\boxed{\delta = \bar{\Pi}_0 = -\bar{\Pi}_1 \bar{+} \bar{\Pi}_2 \bar{+} \bar{G}}$$

$$\Pi_2 = P_c \cdot A_2$$

$$\Pi_1 = (P_c + h^* \gamma_2) A_1.$$

$$G = \gamma_2 W_2$$



$Q_1, Q_2, Q_3.$

$P_r.$ SDB.

Q_3

1) $$z_1 + \frac{v_1^2}{2g} + \frac{p_1}{\gamma} = z_2 + \frac{v_2^2}{2g} + \frac{p_2}{\gamma} \Rightarrow \frac{Q_3}{2g}\left(\frac{1}{D_2^2} - \frac{1}{A_1^2}\right) = \Delta\frac{\gamma_м - \gamma}{\gamma}$$

2) $Q_1 = Q_3 + Q_2$. $\quad f(Q_1, Q_2)$

3) B. tra A e M.

$$z_A \pm 0{,}5\frac{Q_1^2}{2gA_1^2} + l_1 J_1 + \frac{Q_1^2}{2gA_1^2} + 0{,}5\frac{Q_2^2}{2gA_2^2} + l_2 J_2 + \frac{Q_2^2}{2gA_2^2} + z_M +$$

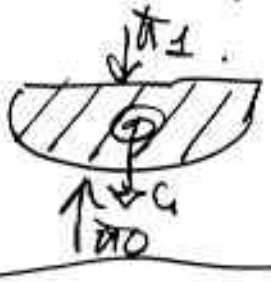
$$+ \Delta Y_m - Q\,r \qquad f(Q_1, Q_2)$$

4) B. tra A e C.

$$\Delta H_p = z_C - z_A + (1+0{,}5)\frac{Q_1^2}{2gA_1^2} + Q_1 J_1 + \frac{Q_3^2}{2gA_3^2} \cdot 0{,}5 + l_3 J_3 +$$

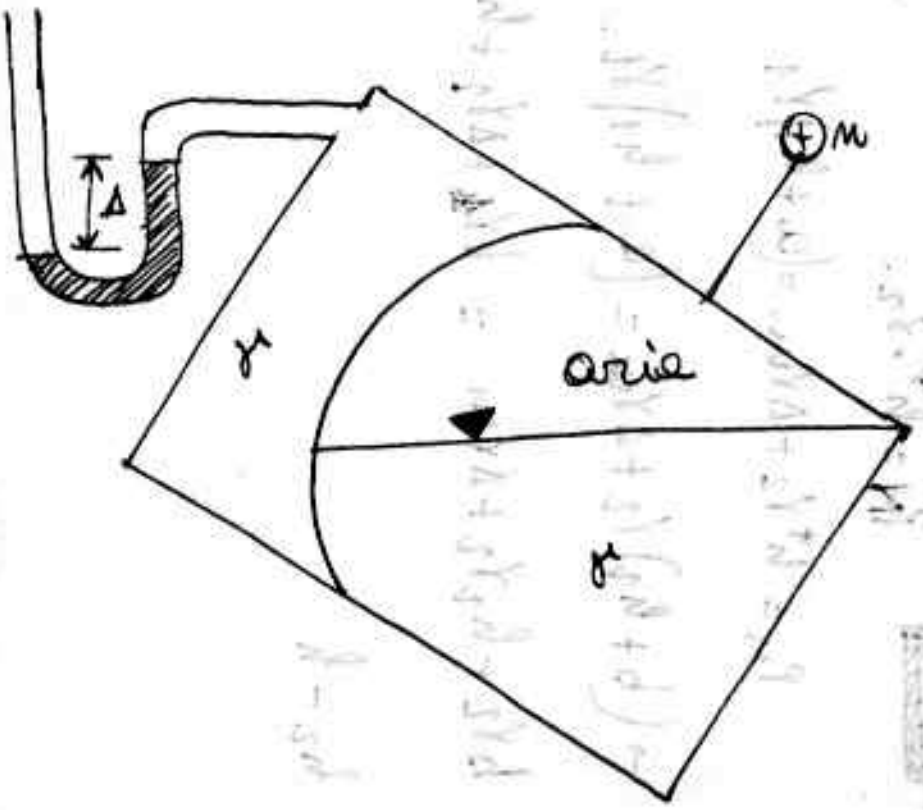
$$+ \frac{Q_3^2}{2g}\left(\frac{1}{A_u} - \frac{1}{A_c}\right)^2 \cdot m \qquad \Rightarrow \Delta H_p .$$

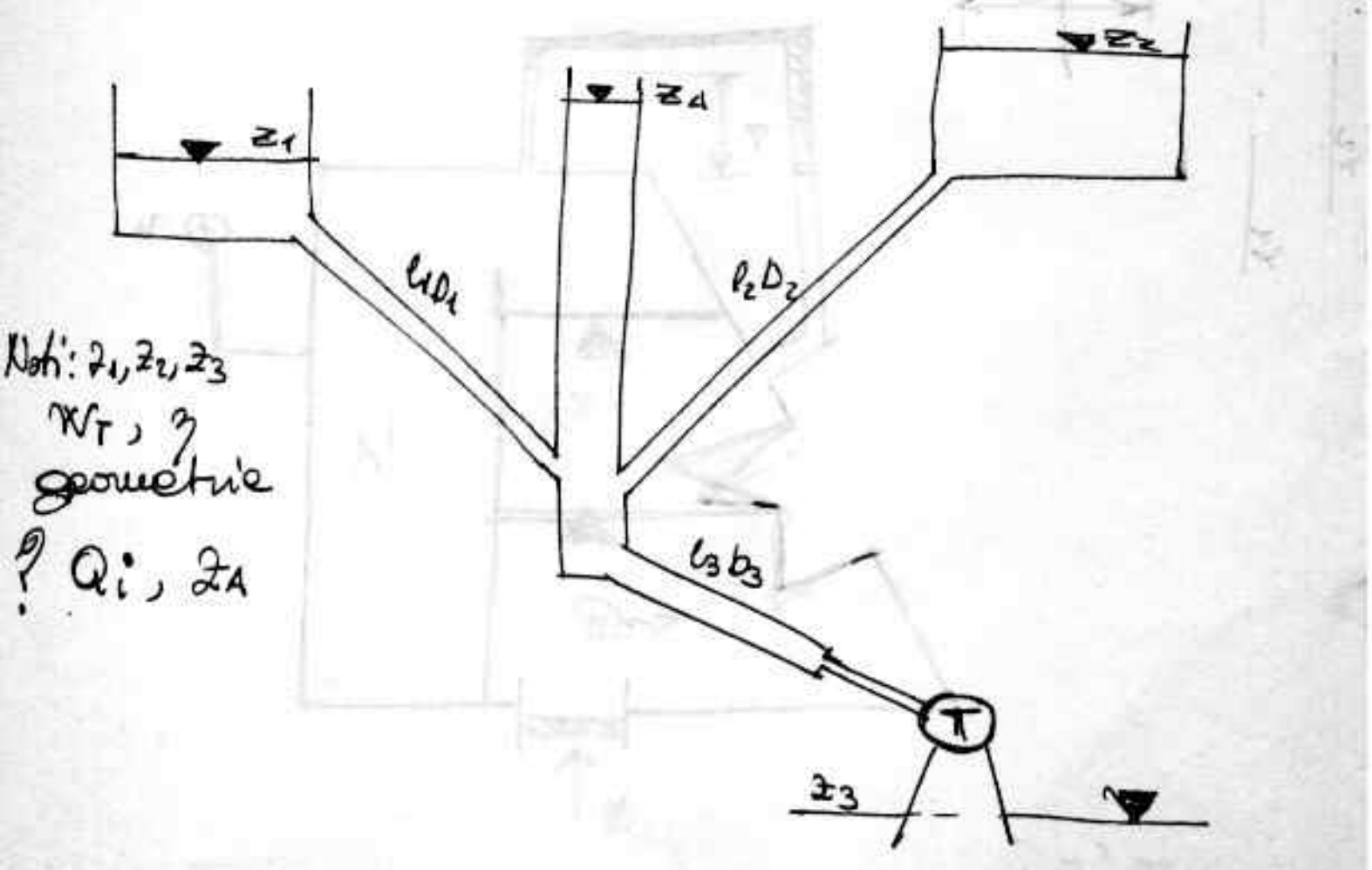
$$P_f = \frac{\gamma Q_3 \Delta H_p}{\eta \cdot 102} \ [kW] .$$



Noti: $\Delta, \gamma, \gamma_m, n$

? Spinta su la calotta sferica

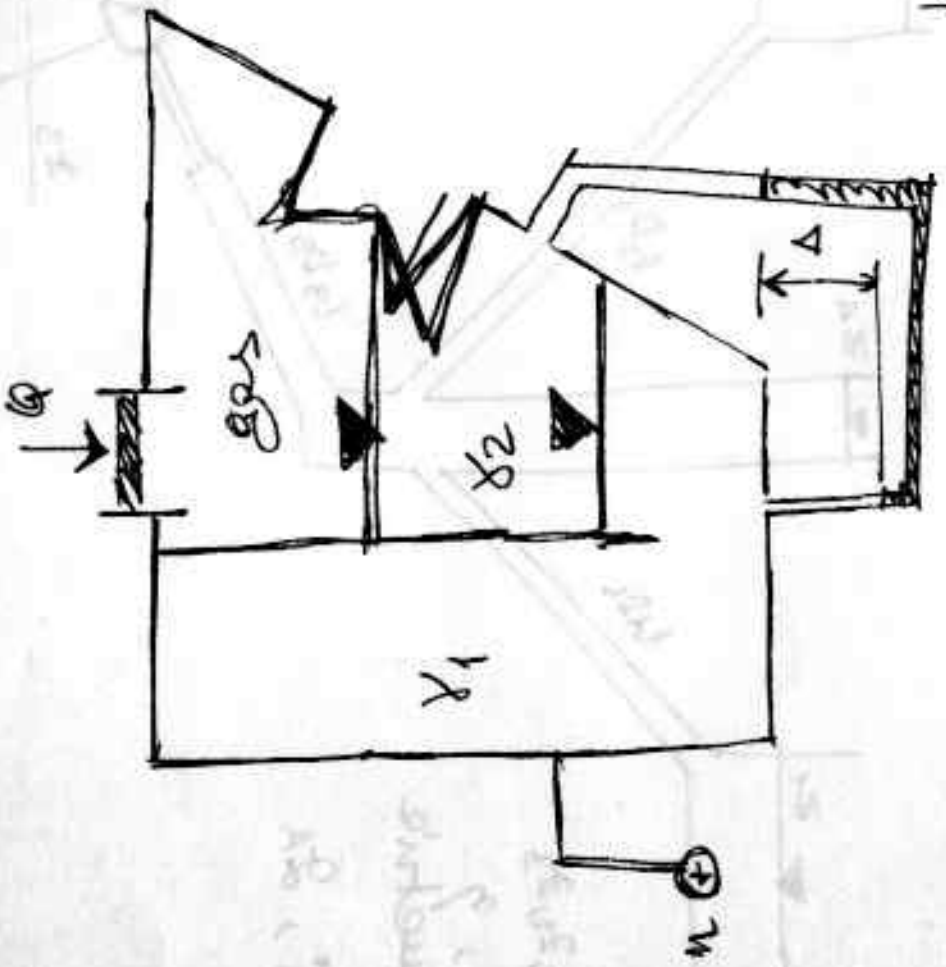


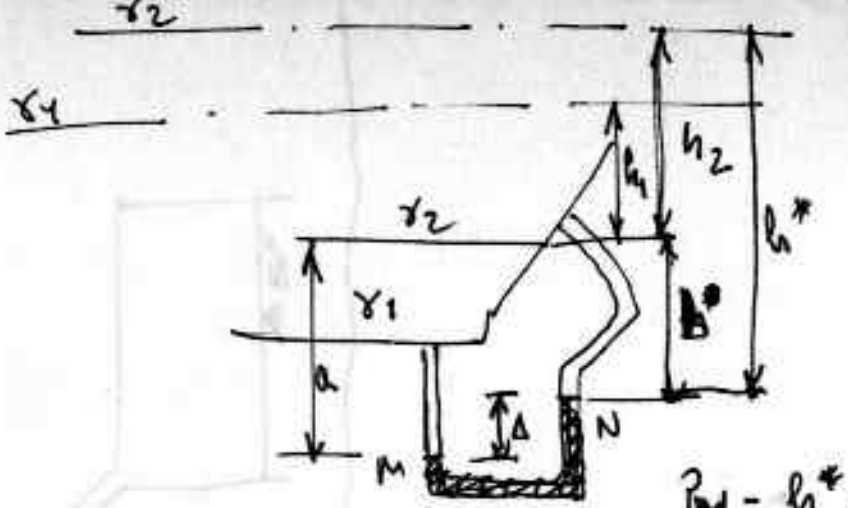


Noti: z_1, z_2, z_3
W_T, 3 geometrie

? Q_i, z_A

q, S, Δ





$P_M = h^* \cdot \gamma_2$.

$P_M = h^* \gamma_2 + \Delta\gamma_{m.} = (a + h_1)\gamma_1$

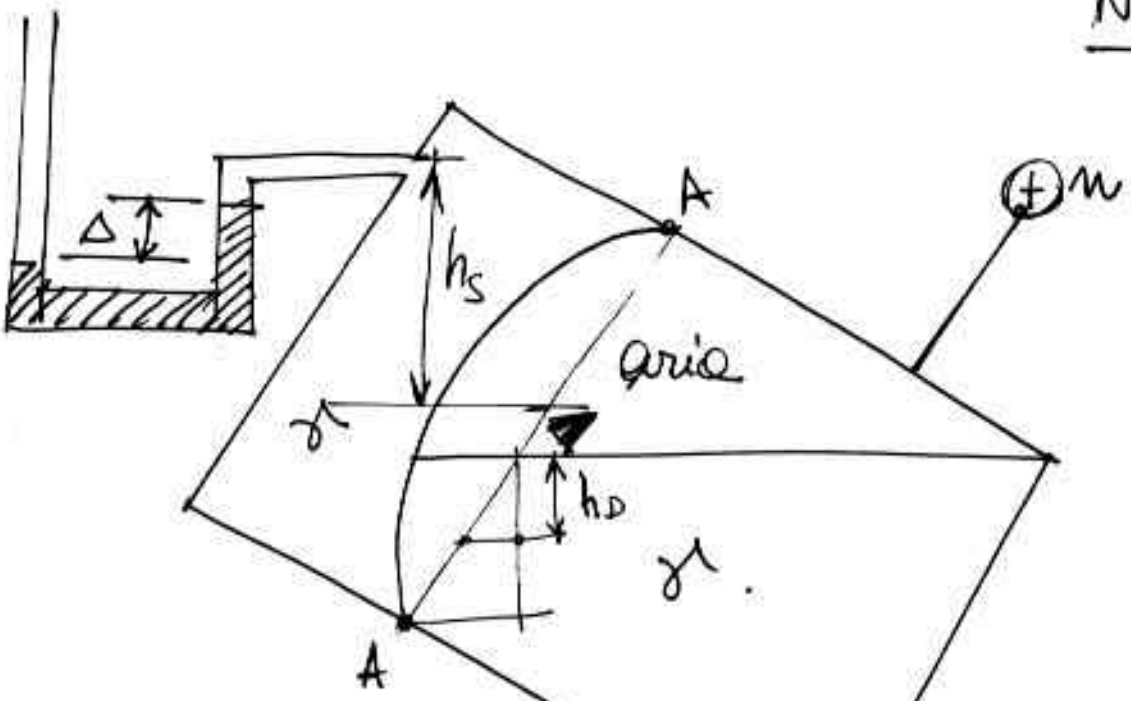
$\rightarrow (b + h_2)\gamma_2 + \Delta\gamma_m = (a + h_1)\gamma_1$

$b\gamma_2 + h_2\gamma_2 + \Delta\gamma_m = b\gamma_1 + \Delta\gamma_1 + h_1\gamma_1$.

$h_2 - h$

Noti: $\Delta, \gamma, \gamma_m, n$

? Spinta su AA



1) Spinta S_1 da sinistra:

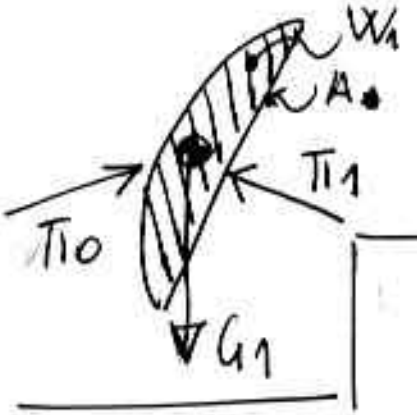
$\bar{\Pi}_0 + \bar{\Pi}_1 + \bar{G}_1 = 0$

$\bar{S}_1 = \bar{\Pi}_0 = -\bar{\Pi}_1 - \bar{G}_1$

$\Pi_1 = (-\Delta\gamma_m + h_S\gamma)A$

dove: ~~$G_1 = (h_S \gamma) A$~~

$G_1 = W_1 \gamma$



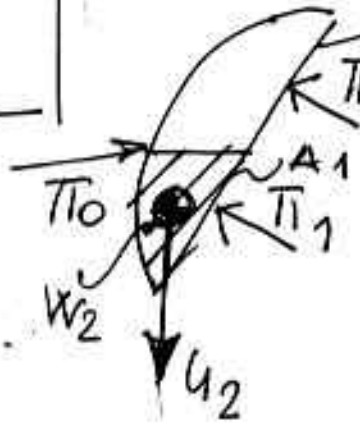
2) Spinta S_2 da destra:

$\bar{S}_2 = -\bar{\Pi}_0 = \bar{\Pi}_1 + \bar{\Pi}_2 + \bar{G}_2$

dove: $G_2 = W_2 \gamma_2$

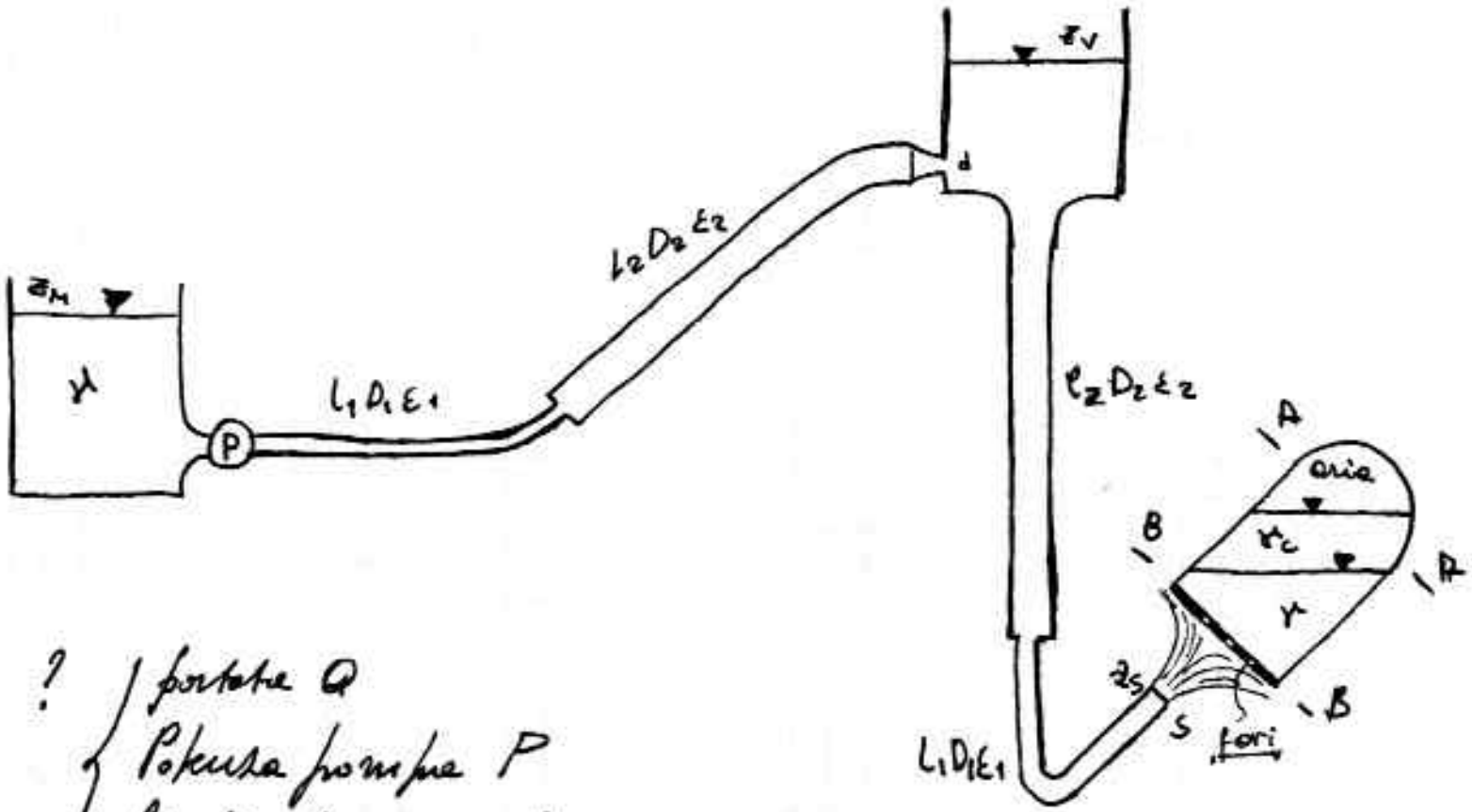
$\Pi_2 = n \cdot 10^4 \cdot A_2$

$\Pi_1 = (n \cdot 10^4 + h_D \gamma) A_1$



ma viene: $\bar{S}_{tot} = \bar{S}_1 + \bar{S}_2$



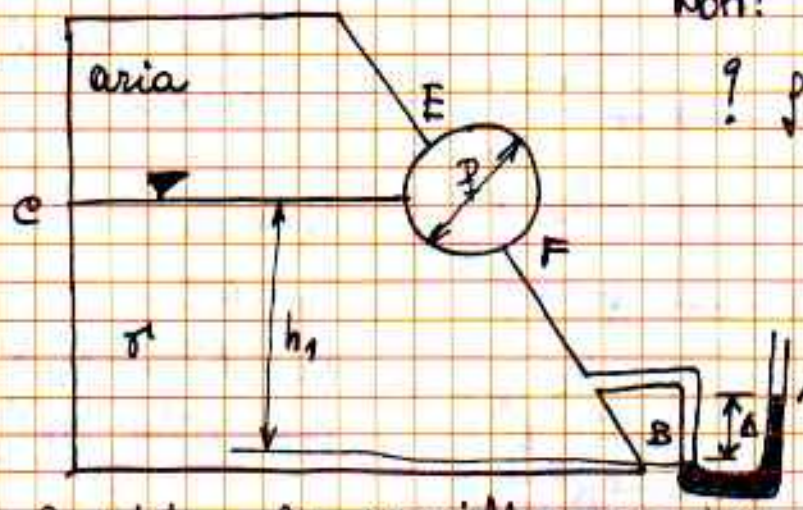


? { portata Q
Potenza pompa P
livello z_V a regime.

? { Spinta totale su BB
Spinta su AA

Noti: γ, γ_m, Δ, la geometria.

? Spinta sulla sfera.

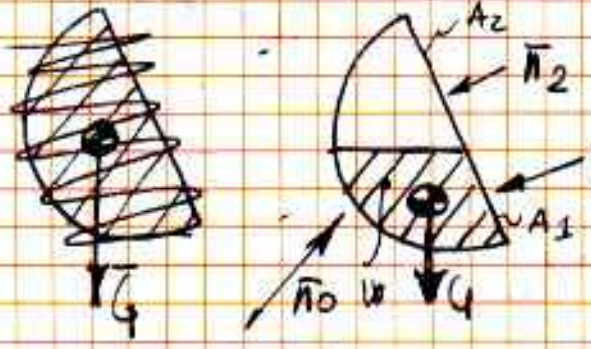


p_B = pabs.

$p_B = \Delta \gamma_m$.

$p_C = \Delta \gamma_m - h_1 \gamma$

Considero la semisfera EF piena di liquido sino al livello coincidente nel serbatoio ed applico l'equazione globale della statica:



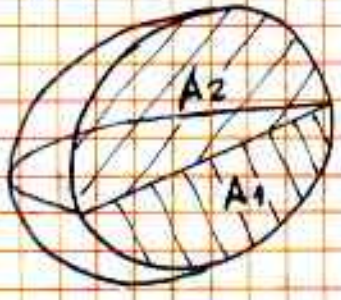
$\bar{\pi}_0 + \bar{\pi}_1 + \bar{\pi}_2 + \bar{G} = 0$.

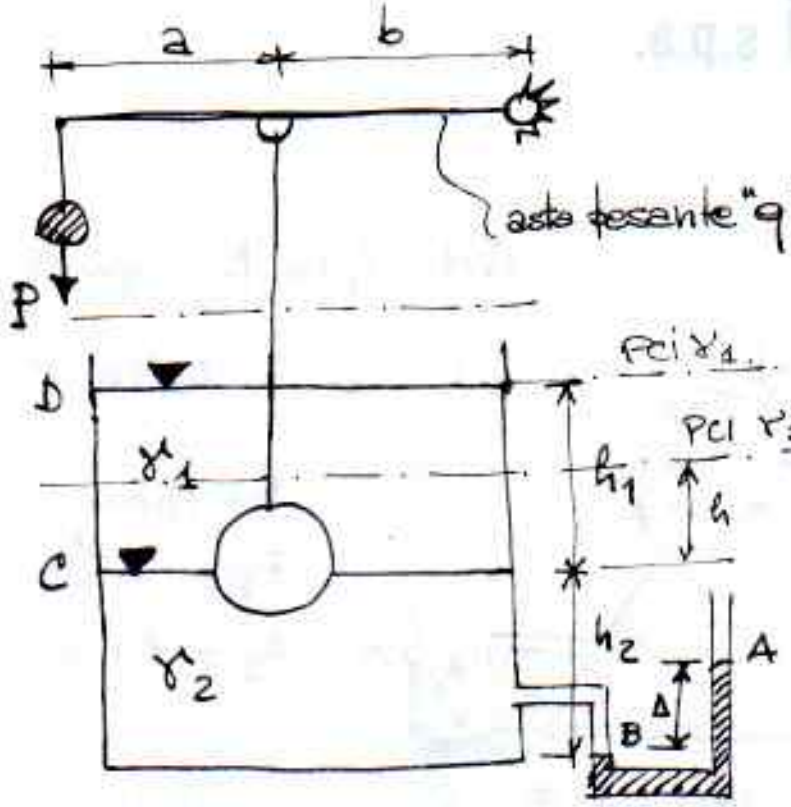
$\bar{S} = \bar{\pi}_0 = -\bar{\pi}_1 - \bar{\pi}_2 - \bar{G}$

dove: $\bar{\pi}_2 = p_C \cdot A_2$

$\bar{\pi}_1 = (h\gamma + p_C) A_2$.

h = affondamento del baricentro dell'area A_2 sotto al pelo libero del liquido.





? P per l'equilibrio.

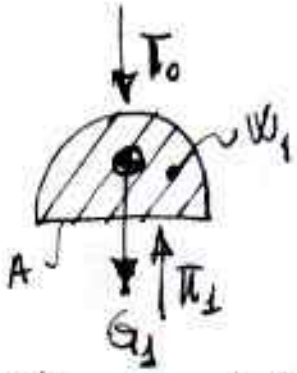
$P_A = p_{atm}$.

$P_B = \Delta\gamma_m$.

$P_C = \Delta\gamma_m - h_2\gamma$

$h = \dfrac{\Delta\gamma_m - h_2\gamma}{\gamma}$

1) Spinta del liquido γ_1.



$\bar{S}_1 = \bar{\Pi}_0 = -\bar{\Pi}_1 - \bar{G}_1$

$\Pi_1 = P_C \cdot A$

$G_1 = \gamma_1 W_1$.

2) Spinta del liquido γ_2.



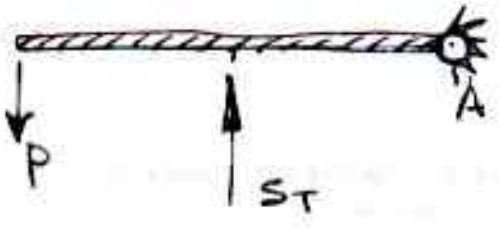
$\bar{S}_2 = \bar{\Pi}_0 = -\bar{\Pi}_1 - \bar{G}_2$

$\Pi_1 = P_C \cdot A$

$G_2 = \gamma_2 W_2$

Spinta totale $\vec{S}_{TOT} = \vec{S}_1 + \vec{S}_2$ ed è diretta verso l'alto.

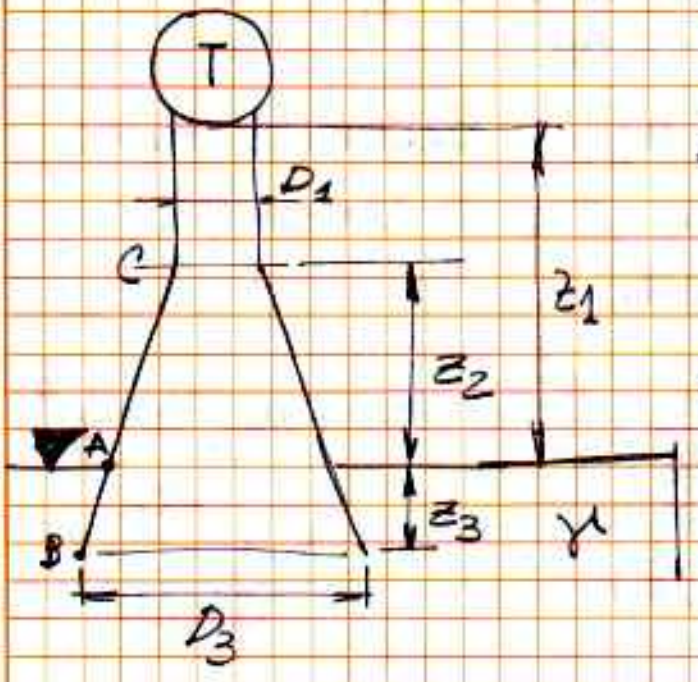
Equilibrio dell'asta: (Momento attorno ad A):



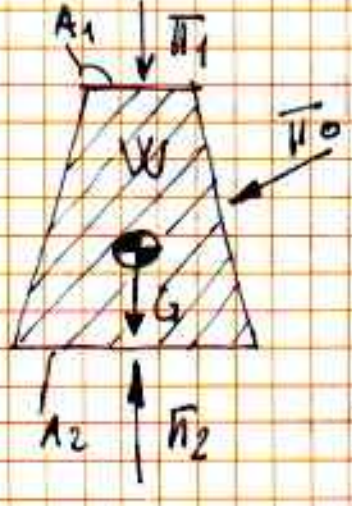
$$P(a+b) + qb = S_T \cdot b.$$

$$P = \frac{b(S_T - q)}{a+b}.$$

1) Yu condiz. statiche
? Spinta sul diffusore.
2) Nota la portata Q a regime
? Spinta totale sul diffusore.
3) ? portata limite Qlim.



1) $p_3 = z_3 \gamma'$, $p_2 = -\gamma z_2$ applico l'eq. globale della statica tra le sezioni A e C.
Sulla parete AB la spinta risultante è nulla, in quanto insiste eq. di pressioni tra interno ed esterno.



$$\bar{\Pi}_0 + \bar{\Pi}_1 + \bar{\Pi}_2 + \bar{G} = 0$$

$$\bar{S} = -\bar{\Pi}_0 = \bar{\Pi}_1 + \bar{\Pi}_2 + \bar{G}$$

dove: $G = \gamma' W$

$\bar{\Pi}_1 = -\gamma z_2 \cdot A_1$

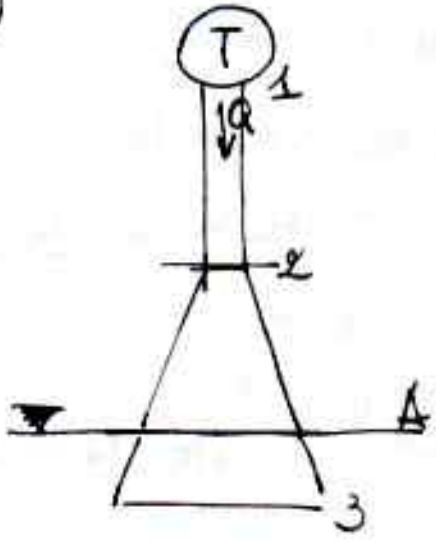
$\bar{\Pi}_2 = 0$ quindi:

$$|\bar{S}| = \gamma' W - \gamma z_2 A_1$$

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Tale spinta è diretta verso l'alto e fissa per l'asse del diffusore.

2)



Nota la portata calcolo le velocità:

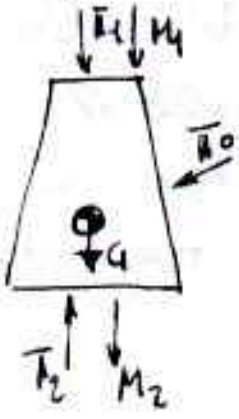
$$V_2 = \frac{Q}{A_2} \qquad V_3 = \frac{Q}{A_3}$$

Quindi applico Bernoulli tra la sezione 2 e il serbatoio di valle.

$$z_2 + \frac{V_2^2}{2g} + \frac{p_2}{\gamma} = z_a + \underbrace{\frac{(V_2 - V_3)^2}{2g}}_{\text{GIBSON}}$$

da tale equazione ricavo p_2

a) Spinta idrodinamica:



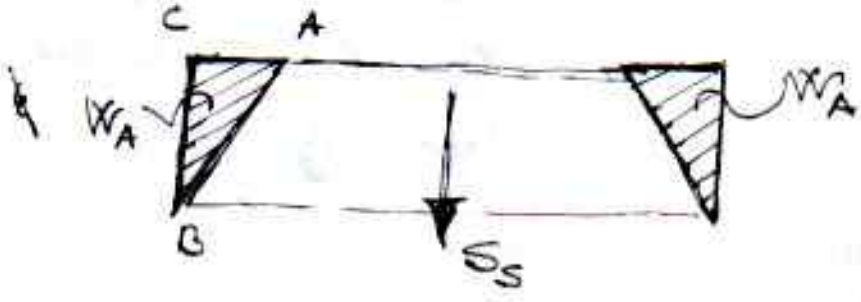
$$\bar{S}_D = -\bar{\Pi}_0 = \bar{\Pi}_1 + \bar{M}_1 + \bar{\Pi}_2 - \bar{M}_2$$

$$\bar{\Pi}_1 = p_2 \frac{D_2^2}{4}\pi \qquad \bar{\Pi}_1 = p_a \gamma z_3 \frac{D_3^2}{4}\pi$$

$$M_1 = \rho V_2 Q \qquad M_2 = \rho V_2 Q \quad (\beta \simeq 1)$$

diretta verso l'alto.

b) Spinta idrostatica è pari al peso del volume di fluido ABC $S_s = G W_A$ è diretta verso il basso. $\bar{S}_{TOT} = \bar{S}_D + \bar{S}_T$



3) Portata limite: si realizza quando in A si ha pressione assoluta nulla.
Applicando il Th. di Bernoulli generalizzato tra 1 e A:

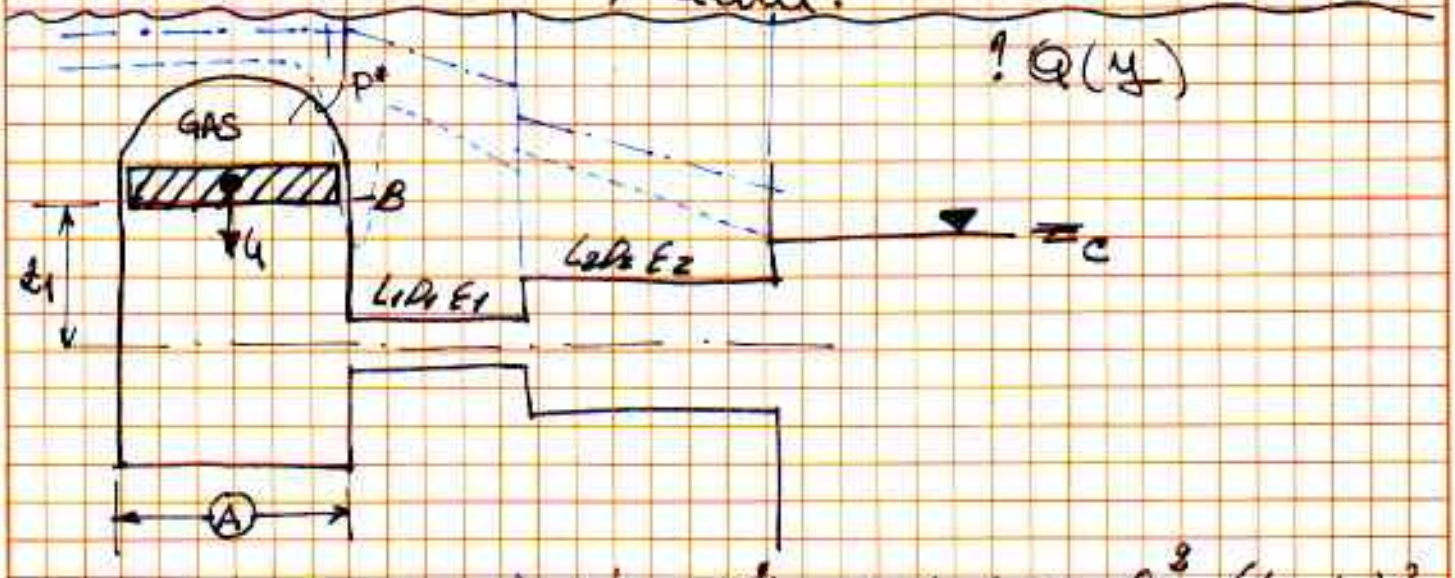
$$z_1 - \frac{p_A^*}{\gamma} + \frac{V_1^2}{2g} = z_A + \frac{(V_2 - V_3)^2}{2g}\,m$$

(Gibson)

esprimendo le velocità in funzione della portata:

$$\frac{Q_{lim}^2}{2g}\left[\frac{1}{A_1} - \left(\frac{1}{A_2} - \frac{1}{A_3}\right)^2 m\right] = z_A - z_1 + \frac{p_A^*}{\gamma}$$

$\rightarrow Q_{lim}$.



$$p_B = p^* + \frac{\gamma}{A}$$

$$z_1 + \frac{p_B}{\gamma} + \frac{V_B^2}{2g} = L_1 J_1 + L_2 J_2 + \frac{Q^2}{2g}\left(\frac{1}{A_1} - \frac{1}{A_2}\right)^2 + 0{,}5\frac{Q^2}{2gA_1} + \frac{Q^2}{2gA_2} + z_c.$$

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$$z_1 + \frac{p_B}{\gamma} + \frac{Q^2}{2gA} = 0{,}5\frac{Q^2}{2gA_1} + L_1 J_1 + \frac{Q^2}{2g}\left(\frac{1}{A_1} - \frac{1}{A_2}\right)^2 + L_2 J_2 + \frac{Q^2}{2gA_2} + z_c .$$

dove : $z_1 - z_c = y$.

$$J_i = \lambda_i \frac{Q^2}{2g D_i A_i^2} \qquad \frac{1}{\sqrt{\lambda_i}} = -2\log\left(\frac{1}{3{,}71}\frac{\varepsilon_i}{D_i}\right)$$